

Lecture Plan

Name of Course	Master of Computer Applications	
AKTU Course Code	09	
Branch	MCA	
Semester	2nd	
Section	A	
Total Number of Students	26	
Name of Faculty	Mr. Arun Kumar	
Number of Lecture Proposed	56	
Subject Name	Data Structures & Analysis of Algorithms	Subject Code: BMC 205

S.No	Unit No	Topic	CO	No of Lectures Required	Actual Date of Completion	Suggested Reference
1	1	Data, Entity, Information, Difference b/w Data & Information, Data type, Build in data type, Abstract data type	1	1	28-01-26	
2		Definition of data structures, Types of Data Structures: Linear and Non-Linear Data Structure,		1	29-01-26	
3		Introduction to Algorithms: Definition of Algorithms, Difference between algorithm and programs, properties of algorithm, Algorithm Design Techniques		1	31-01-26	
4		Performance Analysis of Algorithms, Complexity of various code structures		1	02-02-26	
5		Order of Growth, Asymptotic Notations.		2	04-02-26	
6		Definition, Single and Multidimensional Arrays, Representation of Arrays: Row Major Order, and Column Major Order		1	05-02-26	
7		Derivation of Index Formulae for 1-D,2-D Array Application of arrays, Sparse Matrices and their representations		2	09-02-26	
8		Array Implementation and Pointer Implementation of Singly Linked Lists, Doubly Linked List, Circularly Linked List, Operations on a Linked List. Insertion, Deletion, Traversal, Polynomial Representation and Addition Subtraction & Multiplications of Single variable		3	12-02-26	
No of Lectures Required to complete Unit 1				12		
9		Abstract Data Type, Primitive Stack operations: Push & Pop, Array and Linked Implementation of Stack in C,		2	16-02-26	
10		Application of stack: Prefix and Postfix Expressions. Evaluation of		2	18-02-26	

		postfix expression,	2			
11	2	Iteration and Recursion- Principles of recursion, Tail recursion, Removal of recursion Problem solving using iteration and recursion with examples such as binary search		1	19-02-26	
12		Fibonacci numbers, and Hanoi towers.		1	21-02-26	
13		Operations on Queue: Create, Add, Delete, Full and Empty,		1	23-02-26	
14		Circular queues, Array and linked implementation of queues in C, Dequeue and Priority Queue		1	24-02-26	
15		Concept of Searching, Sequential search, Index Sequential Search,		1	25-02-26	
16		Binary Search.		1	26-02-26	
17		Concept of Hashing & Collision resolution Techniques used in Hashing		1	28-02-26	
No of Lectures Required to complete Unit 2				11		
18	3	Insertion Sort,	3	1	07-03-26	
19		Selection Sort, Bubble Sort,		1	09-03-26	
20		Heap Sort,		2	11-03-26	
21		Comparison of Sorting Algorithms,		1	16-03-26	
22		Sorting in Linear Time: Counting Sort and Bucket Sort		1	17-03-26	
23		Terminology used with Graph		1	18-03-26	
24		Data Structure for Graph Representations: Adjacency Matrices, Adjacency List,		1	19-03-26	
25		Graph Traversal: Depth First Search and Breadth First Search,		2	24-03-26	
26		Connected Component		1	25-03-26	
No of Lectures Required to complete Unit 3				11		
27	4	Basic terminology used with Tree, Binary Trees	4	1	28-03-26	
28		Binary Tree Representation: Array Representation and Pointer (Linked List) Representation		1	30-03-26	
29		Binary Search Tree, Complete Binary Tree, Extended Binary Trees		2	01-04-26	
30		Tree Traversal algorithms: Inorder, Preorder and Postorder		1	02-04-26	
31		Constructing Binary Tree from given Tree Traversal		1	04-04-26	
32		Operation of Insertion, Deletion, Searching & Modification of data in Binary Search Tree.		2	07-04-26	
33		Threaded Binary trees		1	08-04-26	
34		Huffman coding using Binary Tree,		1	09-04-26	
35		AVL Tree		2	13-04-26	
36		B Tree		1	15-04-26	

No of Lectures Required to complete Unit 4				13		
37	5	Divide and Conquer with Examples Such as Merge Sort, Quick Sort,	5	2	18-04-26	
38		Matrix Multiplication: Strassen's Algorithm		1	20-04-26	
39		Dynamic Programming: Dijkstra Algorithm, Bellman Ford Algorithm		2	22-04-26	
40		All- pair Shortest Path: Warshal Algorithm		1	23-04-26	
41		Longest Common Sub-sequence		1	25-04-26	
42		Greedy Programming: Prim's and Kruskal algorithm.		2	28-04-26	
43						
44						
No of Lectures Required to complete Unit 5				9		

Text Books & References	
1	Cormen T. H., Leiserson C. E., Rivest R. L. and Stein C., "Introduction to Algorithms", PHI
2	Horowitz E., Sahni S. and Rajasekharan S., "Fundamentals of Computer Algorithms", Universities Press
3	Aho A., Ullman J. and Hopcroft J., "Design and Analysis of algorithms", Pearson Education
4	Goyal K. K., Sharma S. and Gupta A., "Data Structures and Analysis of Algorithms", HP Hamilton
5	Dave P. H. and Dave H. B, "Design and Analysis of Algorithms", Pearson Education

Course Outcomes (COs)		
At the end of this course students will demonstrate the ability to:		
CO 1	Concept of data structure, abstract data types, algorithms, analysis of algorithms and basic data organization schemes such as arrays and linked lists	K2
CO 2	The applications of stacks and queues and implement various operations on them using arrays and linked lists.	K3
CO 3	The properties of graphs and trees and implement various operations such as searching and traversal on them.	K3
CO 4	Compare incremental and divide-and-conquer approaches of designing algorithms for problems such as sorting and searching.	K4
CO 5	Apply and analyze various design approaches such as Divide-and Conquer, greedy and dynamic for problem solving.	K4

Name of Faculty with Signature: Mr. Arun Kumar

Lecture Plan In-charge: Mr. Praveen Kr. Tripathi

Head of Department: Mr. Prashant Shrivastava

Dean Academics: Dr. Kamal Prakash Pandey